

The Geometry of Validity

What a Reasoning Chain’s Trajectory Shows That a Probe Does Not

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July 30, 2026

Abstract

Large language models often produce reasoning that looks convincing but is invalid, and their written explanations are unreliable witnesses of what they actually compute. Existing work shows that the truth of a single statement can be read from a model’s internal state with a linear probe. A reasoning chain is not a statement but a sequence, and as a model reads one its internal state traces a path through the network’s layers, so we measure the shape of that path with the tools of curve geometry. Valid chains trace smoother paths than broken ones in almost every model and dataset we tested, and the effect survives strict controls for text length, which otherwise imitates geometric signal. Linear readability extends from facts to inference, since a probe on a single pooled internal state detects the validity of an entire chain. Against that probe, trajectory geometry never significantly improves detection. Against a linear compression of the representation to the same number of dimensions it wins, so the redundancy lies in the ceiling rather than in the content. Sensitivity also depends on how a chain was broken, responding to corrupted content and remaining at chance on corrupted order, so the shape of a failure carries information about its kind that a detection score does not. Reasoning validity is written into the curve geometry of a model’s computation, but the model compresses that same information into the linear geometry of every single state.

1 Introduction

Reasoning is not a point but a process. When a language model reads a multi-step reasoning chain, its hidden states trace a discrete path through representation space, across token positions within a layer and across layers in depth. If the truth of a single statement is linearly encoded, the question that follows is whether the validity of a multi-step chain is reflected in the shape of that path. A statement’s factuality is a property of a point, whereas a chain’s validity is a property of a path, and the natural language for paths is the differential geometry of curves [Do Carmo, 2016].

We are interested in this question for two reasons. Practically, verifying multi-step reasoning is a central open problem [Lightman et al., 2024], and chain-of-thought text is an unreliable witness of the underlying computation [Turpin et al., 2023, Lanham et al., 2023]. Theoretically, a substantial literature studies the static geometry of language model representations (§2), but whether the trajectory of computation carries interpretable structure of its own is largely unexplored, and an earlier version of this work, along with related physics-inspired analyses [Greydanus et al., 2019, Amari, 2016], conjectured that it does.

We define discrete estimators of path length, curvature, torsion and straightness on hidden-state trajectories, prove their basic properties in a proof assistant, and evaluate them against two baselines on nine independent model and dataset combinations. Our findings are the following.

Valid chains trace smoother paths through depth. Layer curvature is significantly lower for valid chains in eight of nine runs and layer torsion in seven of nine, with the same sign everywhere, after two independent controls for text length.

Linear readability extends from facts to inference. A probe on one mean-pooled hidden state detects the validity of a complete chain, not only the truth of a single statement, at up to 0.99 AUROC.

As a detector, trajectory geometry is redundant. Stacking the geometric features onto that probe never significantly improves detection, and significantly degrades it in two runs.

At equal dimensionality it is not redundant. Against a linear compression of the pooled state to the same size, the geometric features win in seven of nine runs, which places the limitation in the ceiling of a low-dimensional summary rather than in the content of the geometry.

Trajectory shape reads content and ignores order. Scored against each way of breaking a chain, the geometric detector responds to substituted premises and conclusions and sits at chance on shuffled steps, whereas the probe responds to all three.

2 Related work

Probing truth-like properties. Supervised linear probes [Azaria and Mitchell, 2023, Levinstein and Herrmann, 2023], unsupervised contrast-consistent search [Burns et al., 2022] and representation-engineering approaches [Zou et al., 2023] train probes on internal activations to classify truthfulness. Marks and Tegmark [2023] consolidated this line, showing that difference-in-means directions generalize well and are causally implicated, while Levinstein and Herrmann [2023] documented generalization failures of earlier probes. All rely on a single representation per input. We adopt the linear probe on one pooled hidden state as our baseline and ask whether the trajectory adds anything to it. Model self-knowledge of correctness, studied through calibrated uncertainty [Kadavath et al., 2022] and semantic entropy [Farquhar et al., 2024], motivates our second baseline family.

Static geometry of LLM representations. Contextual embedding spaces are anisotropic [Ethayarajh, 2019] with cluster and manifold structure [Cai et al., 2021], intrinsic dimension varies systematically across layers [Valeriani et al., 2023, Aghajanyan et al., 2021], and many semantic features are linearly represented [Park et al., 2023, Nanda et al., 2023, Tigges et al., 2024]. Depth-wise analyses such as the tuned lens [Belrose et al., 2023] treat the layer stack as a refinement process, which motivates our layer-trajectory construction. Physics-inspired formalisms have already been applied to neural computation, for example Hamiltonian neural networks [Greydanus et al., 2019] and information geometry [Amari, 2016].

Reasoning chains and their verification. Chain-of-thought prompting draws out multi-step reasoning [Wei et al., 2022, Kojima et al., 2022], but generated rationales can be unfaithful to the underlying computation [Turpin et al., 2023, Lanham et al., 2023], which motivates process-level verification [Lightman et al., 2024]. Reasoning in continuous latent spaces [Hao et al., 2024] makes trajectory-level analysis natural. Our three corpora with validity labels are listed in Table 1 [Mihaylov et al., 2018, Jhamtani and Clark, 2020, Dalvi et al., 2021, Saparov and He, 2022].

3 Discrete geometric estimators

Trajectories. A transformer [Vaswani et al., 2017] with L layers and hidden width h turns every token of an input text into one vector of dimension h per layer. From these vectors we build two ordered lists of points for each reasoning chain, and each such list is a trajectory.

The token trajectory follows the text. We fix a layer ℓ^* at two thirds of the network’s depth, chosen before any experiment and consistent with prior probing work [Marks and Tegmark, 2023], and collect the hidden state of every token position there, giving $X^{\text{tok}} = (x_1, \dots, x_n)$. The layer trajectory follows depth instead. For each layer ℓ we average the hidden states of all token positions into a point $\bar{x}_\ell$, and $X^{\text{lay}} = (\bar{x}_0, \dots, \bar{x}_L)$ describes how the representation of the whole chain is refined from the first layer to the last.

All curve-geometry quantities are computed in the full dimension h , since the quantities defined below depend only on distances and inner products between points and need no dimensionality reduction.

Global shape. We capture the coarse shape of a trajectory $X = (x_1, \dots, x_n)$ with three measures. Writing $\Delta_i = x_{i+1} - x_i$ for the step from one point to the next, the path length $S(X) = \sum_i \|\Delta_i\|$ is the total distance travelled, the Dirichlet energy $E(X) = \frac{1}{2} \sum_i \|\Delta_i\|^2$ measures the same path but penalizes large jumps more heavily than small ones, and the straightness $\sigma(X) = \|x_n - x_1\|/S(X)$ compares the direct distance between the endpoints with the distance actually travelled. A perfectly straight trajectory has straightness equal to one.

Curvature. Curvature measures how sharply the trajectory turns at each interior point. At x_i the trajectory arrives along Δ_{i-1} and leaves along Δ_i , and the angle θ_i between them is how much the direction of travel changed, so $\theta_i = 0$ means the path continues straight ahead. Dividing the angle by the average length of the two steps gives the turn per unit of distance travelled, which is the standard discrete counterpart of the curvature of a smooth curve [Do Carmo, 2016].

$$\kappa_i = \frac{\theta_i}{\bar{s}_i}, \quad \theta_i = \arccos \frac{\langle \Delta_{i-1}, \Delta_i \rangle}{\|\Delta_{i-1}\| \|\Delta_i\|}, \quad \bar{s}_i = \frac{1}{2} (\|\Delta_{i-1}\| + \|\Delta_i\|). \quad (1)$$

Torsion. A trajectory can turn without leaving a flat plane, the way a road curves on a map, and torsion is nonzero only when the path twists out of its own plane, the way a spiral staircase does. Estimating it requires four consecutive points, from which we approximate the first three derivatives of the underlying curve by finite differences, $v_1 = \frac{1}{2}(x_{i+1} - x_{i-1})$, $v_2 = x_{i+1} - 2x_i + x_{i-1}$ and $v_3 = x_{i+2} - 3x_{i+1} + 3x_i - x_{i-1}$. The unsigned torsion is then

$$|\tau_i| = \frac{\sqrt{\det G(v_1, v_2, v_3)}}{\|v_1 \wedge v_2\|^2}, \quad \|v_1 \wedge v_2\|^2 = \|v_1\|^2 \|v_2\|^2 - \langle v_1, v_2 \rangle^2. \quad (2)$$

Here G is the matrix of inner products of the three difference vectors, and its determinant measures the volume they span, which is zero exactly when the trajectory stays in a plane. The denominator normalizes by the area of the plane defined by the first two vectors. All powers of the step size cancel between numerator and denominator, so the estimate does not depend on how densely the curve is sampled.

From trajectories to features. When a step vanishes there is no direction, and when the first two difference vectors are nearly parallel there is no well-defined plane against which to measure twisting. We exclude these points by explicit numerical guards, stated in Appendix A, and the excluded fraction is below 0.001 in every run. Each chain is then summarized by the mean, maximum and standard deviation of its curvatures and torsions, together with S , E , σ and a small set of energy-style summaries retained from the Hamiltonian formalism (Appendix A).

Verification. Estimators of this kind are easy to implement incorrectly, and mistakes are hard to notice because the data offers no ground truth. We therefore validate twice before any model data is processed. The numerical validation recovers the closed-form curvature and torsion of sampled circles, helices and straight lines embedded in high dimension by random rotations, to

Table 1: Reasoning-chain corpora. OpenBookQA carries crowd-sourced validity annotations [Mihaylov et al., 2018, Jhamtani and Clark, 2020]. EntailmentBank [Dalvi et al., 2021] and PrOntoQA [Saparov and He, 2022] are balanced by construction, each valid chain paired with one corruption assigned in equal proportion under a fixed seed.

Dataset	Chains	Valid/invalid	Validity source
OpenBookQA	998	95/903	crowd annotation, majority vote
EntailmentBank	1,200	600/600	gold trees vs. corruptions
PrOntoQA	560	280/280	gold proofs vs. corruptions

four decimal places, and confirms that every feature is unchanged under random rotations and translations (Appendix A). The formal validation states the basic mathematical properties of the estimators as theorems in the Lean 4 proof assistant [Moura and Ullrich, 2021] and has the computer check every step.

Proposition 1 (machine-checked, Appendix B). *For any finite trajectory X in a real inner-product space, (i) $S(X) \geq 0$ and $E(X) \geq 0$, (ii) $\|x_n - x_1\| \leq S(X)$, hence $\sigma(X) \in [0, 1]$, and (iii) S , E and σ are invariant under every isometry of the ambient space, while turning angles in (1) are invariant under linear isometries composed with translations.*

Why this matters. These statements hold for every finite trajectory in every real inner-product space, so they are properties of the definitions rather than findings about our models. Parts (i) and (ii) make straightness a usable scale, so any value outside $[0, 1]$ in our code is necessarily a bug. Part (iii) matters most. Nothing in a transformer fixes an origin, an orientation or a preferred set of axes for the residual stream, so a measurement that changed under rotation, reflection or translation would report partly on an arbitrary coordinate choice, and any association we found with validity could be an artifact of it. Invariance means our features describe the shape of the path and not the frame used to write it down. Appendix B explains why we prove this mechanically rather than on paper.

Scope of the geometric claim. In Klein’s sense a geometry is its invariants under a transformation group, and that is what Proposition 1 establishes. We claim nothing beyond it, and in particular no conservation law, for the reasons set out in Appendix B.

4 Experimental setup

Data and models. Table 1 summarizes the datasets, with corruption construction and label rules pre-specified in Appendix A. Each chain is processed by three open-weight models from different families, including a reasoning-distilled variant, which are Mistral-7B-Instruct-v0.3 [Jiang et al., 2023], Qwen2.5-7B-Instruct [Hui et al., 2024] and DeepSeek-R1-Distill-Llama-8B [Guo et al., 2025], in bfloat16 with a 768-token context and no truncation. This yields nine independent model and dataset runs.

Feature families and baselines. We compare three families, all reading the same internal states of the same models on the same chains under identical cross-validation.

- **Geometric.** The §3 features on both trajectories, twenty-one features after the degeneracy guards of Appendix A remove features that are undefined on a substantial fraction of chains.

- Uncertainty. Chain NLL, mean and maximum next-token entropy, mean attention entropy at ℓ^* , and effective rank [Roy and Vetterli, 2007] of the hidden-state matrix, which together form the model-self-knowledge family [Kadavath et al., 2022, Farquhar et al., 2024].
- Static probe. Logistic and mass-mean linear probes [Alain and Bengio, 2016, Marks and Tegmark, 2023] on the single mean-pooled hidden state at ℓ^* .

Chain length in tokens is included as a nuisance covariate in every classifier so that no family earns credit for measuring length.

Statistical protocol. Per-feature inference uses Welch’s t and Mann–Whitney U [Mann and Whitney, 1947] with Benjamini–Hochberg FDR at $q < 0.05$ [Benjamini and Hochberg, 1995], Cohen’s d , Cliff’s δ [Cliff, 1993] and 10,000-resample bootstrap intervals. Because chain length contaminates trajectory features (§5.5), every per-feature claim must survive two independent length controls, which are residualization on the nonlinear basis $\{1, n, n^2, \log n\}$ and a fully nonparametric Mann–Whitney within length deciles, Stouffer-combined [Whitlock, 2005].

Classification uses class-weighted logistic regression under fivefold stratified cross-validation repeated ten times with fixed seeds. The primary metric is AUROC, since the OpenBookQA base rate is 90.5 per cent and raw accuracy would be uninformative, with MCC and balanced accuracy secondary. Family comparisons use DeLong tests [DeLong et al., 1988] on out-of-fold scores together with label-permutation tests, and families are combined by a cross-validated learned stacker. A negative control ran the identical pipeline on features from a random-weight model and correctly reported chance AUROC and no significant features.

5 Results

5.1 Valid chains trace smoother depth trajectories

Figure 1 shows that the standard deviation of layer curvature is significantly lower for valid chains in eight of nine runs, with length-adjusted d from -0.28 to -0.81 and q down to 1.1×10^{-11} . Mean layer curvature is significant in seven of nine runs and mean layer torsion in seven of nine, reaching $d = -0.85$ for Mistral on OpenBookQA, where the raw unadjusted effect reaches -0.99 . Every significant effect has the same sign across all datasets, models and both control methods, so valid chains bend and twist less as their representation is refined through depth. The single consistent exception is Qwen2.5 on PrOntoQA, where no geometric feature survives either control, and we return to it in §6.

Two cautions belong with these numbers. Our largest effects come from OpenBookQA, our smallest and least balanced corpus at 95 valid chains against 903, with labels from crowd annotation rather than construction. The EntailmentBank runs, balanced at 600 against 600 with gold entailment trees, are the cleaner design and carry effects from -0.31 to -0.52 , so we read EntailmentBank as the primary evidence and OpenBookQA as a noisier check that happens to produce larger effects.

5.2 Detection benchmark

We now benchmark three detectors of invalid reasoning under identical cross-validation. The geometric detector uses the features of §3, the uncertainty detector uses the model’s own confidence signals, and the probe detector uses a linear probe on a single pooled hidden state. We also test combinations through a learned stacker, so that a stronger detector is never penalized by averaging with a weaker one (Appendix C).

The first comparison sets the geometric detector against uncertainty. Adding geometric features to the uncertainty detector improves AUROC significantly on all three EntailmentBank

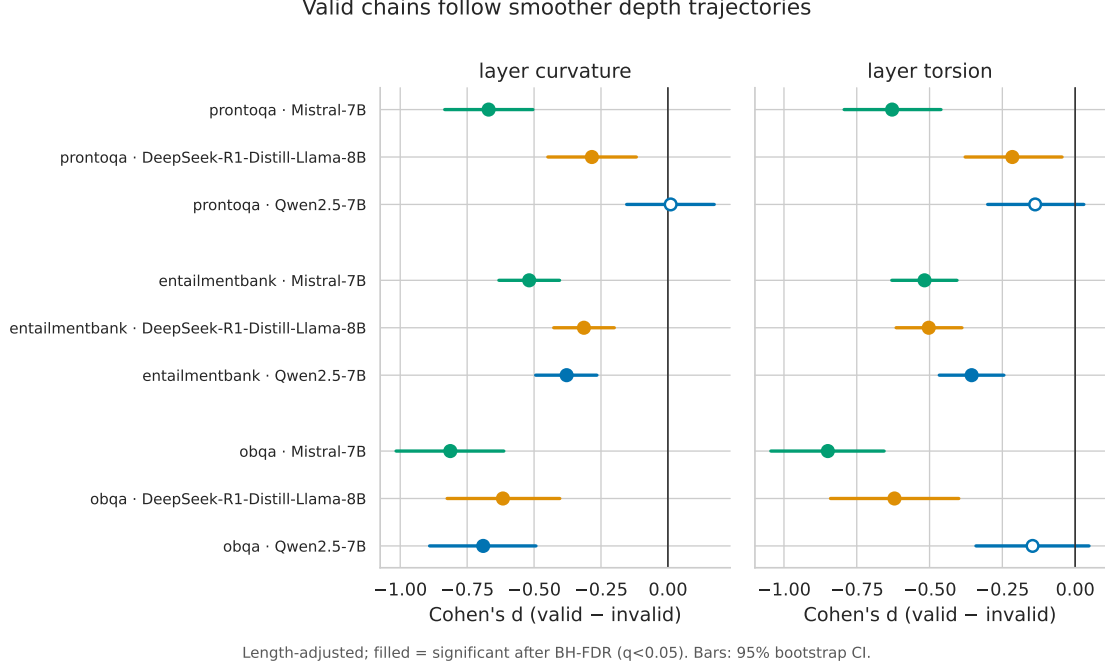


Figure 1: Length-adjusted effect sizes for the two main features across all nine runs. Layer curvature is significantly lower for valid chains in eight of nine runs and layer torsion in seven of nine. Filled markers are significant under BH-FDR ($q < 0.05$) after nonlinear length residualization and also under the stratified test. Bars are 95 per cent bootstrap intervals.

runs, by +0.022 to +0.027 with DeLong $p = 0.003$ to 0.009 and permutation $p \leq 0.011$, and substantially on PrOntoQA, by +0.058 to +0.124. It does not improve OpenBookQA, where the change runs from -0.004 to +0.036 and is significant only for DeepSeek at $p = 0.048$. Trajectory geometry is therefore not reducible to perplexity.

The second comparison sets everything against the probe. The probe alone reaches 0.77 to 0.82 AUROC on OpenBookQA, 0.71 to 0.86 on EntailmentBank and 0.98 to 0.99 on PrOntoQA, dominating both other detectors on every dataset and every model. Stacking the geometric features onto the probe changes AUROC by between -0.007 and +0.012 across the nine runs and is never significantly positive. The best case is Mistral on OpenBookQA at +0.012 with $p = 0.19$. In two runs the stack is significantly worse, which is Qwen on OpenBookQA at -0.007 with $p < 10^{-4}$ and DeepSeek on PrOntoQA at -0.001 with $p = 0.038$.

It follows that the information trajectory geometry carries about validity is already linearly present in a single pooled hidden state. The smoothness signature of §5.1 is a real phenomenon, but as a detector it is a redundant readout of the representation’s content rather than an independent channel.

5.3 Matched baselines

The comparison above is not matched on capacity or on access to depth, and the strength of the conclusion depends on both. We therefore report two additional controls.

Matched dimensionality. Part of the probe’s advantage is capacity rather than information, since it reads h dimensions and the geometric detector reads twenty-one. Compressing the pooled state to twenty-one dimensions before probing weakens it substantially. Against random projections the geometric features win in seven of nine runs, and against a twenty-one component PCA they win on all three EntailmentBank runs. The full-dimensional probe remains strongest

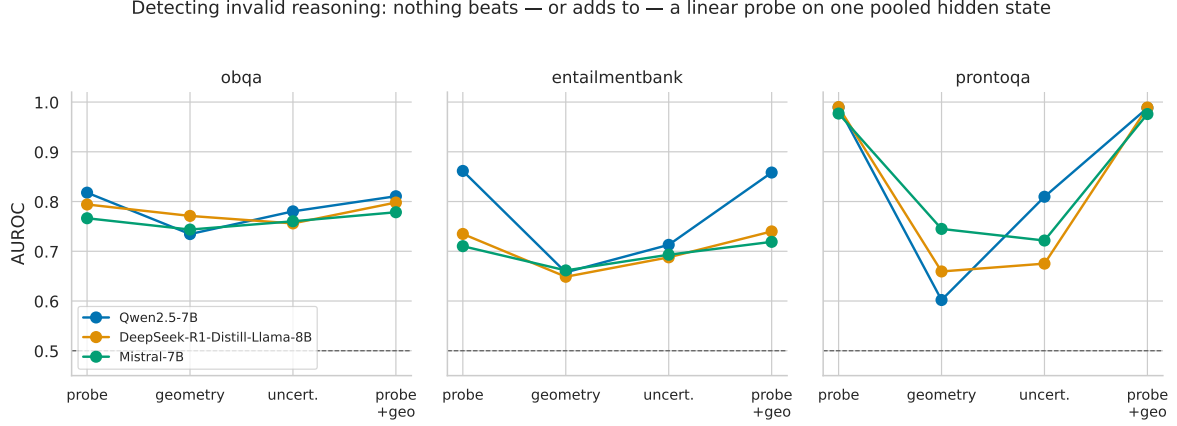


Figure 2: Validity detection by method family, per dataset and model, as out-of-fold AUROC. The probe is the better of the logistic and mass-mean probes on one pooled hidden state [Marks and Tegmark, 2023]. The rightmost method stacks the probe and the geometric detector through a cross-validated combiner. The probe dominates every run and stacking never significantly improves it.

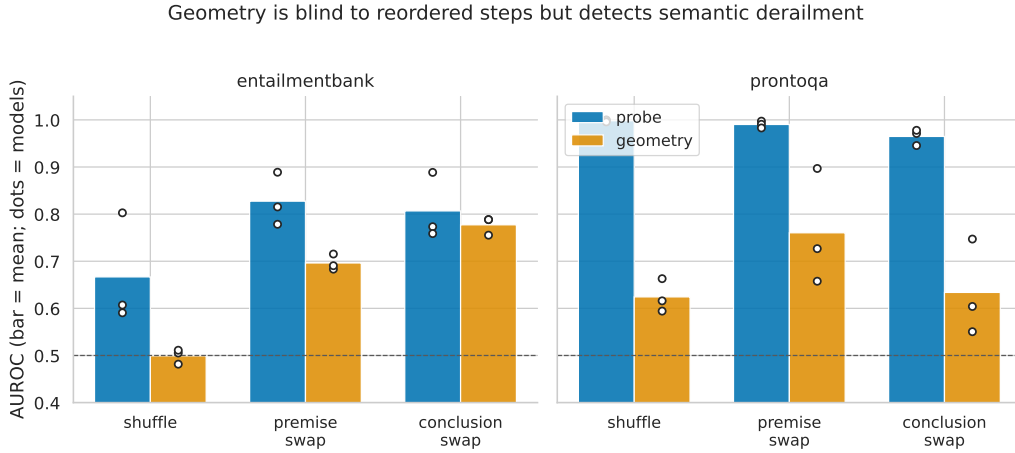


Figure 3: Detection score by type of corruption, comparing valid chains against one corruption class at a time. Bars show the mean over the three models and dots show the individual models.

everywhere, so the conclusion of §5.2 stands, but at equal dimensionality the curve-geometry features are an efficient summary of the validity signal rather than a redundant one.

Matched access to depth. The geometric detector reads all $L + 1$ layers whereas the probe reads one, so a linear reader with the same access to depth is the stronger baseline for a depth-based method. We train the identical logistic probe on the concatenated per-layer pooled states $(\bar{x}_0, \dots, \bar{x}_L)$ under the same cross-validation protocol. [Insert result here before submission. See the comment block in the source for the two outcome branches.]

5.4 Trajectory shape reacts to wrong content and ignores wrong order

Because every invalid chain was produced by a known corruption, we can score each detector separately on each way of breaking a chain. Shuffling reorders the steps but keeps every sentence intact, premise substitution replaces one step with a sentence from another problem, and conclusion substitution replaces the final conclusion. Figure 3 shows the outcome, and it is a

clean split.

The geometric detector is insensitive to shuffled steps. Its mean AUROC on this corruption is 0.55 on EntailmentBank and 0.50 on PrOntoQA, which is chance level, even though shuffling destroys the logic of the chain completely. Against corruptions that change the content the geometric detector works well. It detects premise substitution at 0.69 on EntailmentBank and 0.76 on PrOntoQA, and conclusion substitution at 0.79 on EntailmentBank, close to the probe at 0.81. The probe itself sees all three corruption types, with scores from 0.66 to 0.99.

This split explains the detection benchmark. The shape of the trajectory responds to what the representation contains, and a foreign sentence or a wrong conclusion visibly bends the path through the layers, but it carries no signal about the order in which that content arrives, which the pooled representation still registers weakly. In these models, the curve geometry of the trajectory reads content and not structure. Unlike the association of §5.1, this is a controlled manipulation, since corruption type was assigned by construction under a fixed seed rather than observed.

5.5 The importance of chain length

In raw features, chain length correlates with token-trajectory path length, straightness and energy at $|\rho|$ up to 0.99, and with effective rank at 0.99. Before adjustment, more than a dozen token-level geometric features show significant validity effects that vanish under residualization, because on OpenBookQA valid chains are simply shorter. Even layer-trajectory features, computed on a fixed number of points, inherit length dependence through token pooling at $|\rho|$ up to 0.89. The features surviving both controls in most runs coincide with the layer-smoothness measures of §5.1.

6 Discussion

Each layer of a transformer takes the current representation of the chain and refines it, and when the chain makes sense every refinement pulls in roughly the same direction, so the path through the layers is smooth. We found that a chain containing material that does not belong has a trajectory with higher curvature and higher torsion. Our reading is that successive layers stop agreeing on how the representation should be refined, so the direction of refinement changes from one layer to the next, and accumulated changes of direction are what curvature and torsion measure. The bending of the path is, on this account, the geometric record of a disagreement between layers about what the chain means.

We state this as an interpretation and not as a result, because we have not intervened in the computation to test it. The reading is testable. Decomposing each layer step into its component along the probe direction and its orthogonal complement, and recomputing curvature from each part separately, would show whether the validity signal in curvature lives in the linearly readable subspace, as this account predicts.

On this account, the curve geometry of the trajectory is a shadow of the content the model represents, and this single idea explains our findings. The smoothness effect replicates widely because wherever content is encoded its shadow appears. The probe makes geometry redundant as a detector because it reads the content directly rather than its shadow. The matched-dimensionality result fits too, since a shadow can be a compact and faithful summary while containing no information the original lacks. That a single pooled state supports linear reading of chain validity says that the summary each layer maintains is rich enough to contain the coherence of an entire argument. Even the one exception fits. On the fictional ontologies of PrOntoQA, where world knowledge is useless, the probe still reads validity accurately, so the content is present, but the layer dynamics of Qwen do not visibly bend around it.

Limitations. Our evidence has two levels and lacks a third. The smoothness signature of §5.1 is an observed association, controlled for text length by two independent methods. The dissociation of §5.4 rests on controlled manipulation of the input, so it does tell us what the measurements respond to. What we do not perform are interventions on the model’s internal computation, of the kind used by Marks and Tegmark [2023], so no claim here is causal at the level of mechanism and the layer-disagreement account remains an interpretation. Beyond that, we study three models in the 7 to 8 billion parameter class, English corpora and one pre-specified token layer, with layer sweeps in Appendix C. Invalid chains in EntailmentBank and PrOntoQA are synthetic corruptions, and OpenBookQA labels are crowd annotations on an unbalanced corpus, which is why we treat EntailmentBank as primary. The Lean development verifies the estimators and not the statistical procedures. Finally, our models read the chains rather than generate them, and the trajectories of chains a model produces itself [Wei et al., 2022, Hao et al., 2024] are the immediate extension of this work.

7 Conclusion

Reasoning quality is visible in the shape of a model’s internal trajectories. Valid chains move smoothly through the layers and broken chains bend and twist, in almost every model and dataset we tested, and not as an artifact of text length. We also found its limit. A linear probe on one hidden state detects invalid reasoning at least as well as the full geometric analysis in every single run, so whatever the trajectory knows, that probe already reads.

Detection and interpretation are different problems. The probe answers one question, whether the chain is valid, and returns one number. That is enough for an alarm, but an alarm says nothing about the reasons, and when we want to fix a model rather than flag its output, invalid is not sufficient to guide an intervention.

The curve geometry of the trajectory describes the whole process instead of scoring its outcome, and this difference has three practical consequences. The first is location, since the layer trajectory has one point per layer and we can see at which depth the path starts to bend. The second is diagnosis, since trajectory shape reacts to wrong content and ignores wrong order, so the shape of a failure says something about the kind of failure, which no alarm can do. The third is comparison between models. A probe trained on one model means nothing for another, since each model has its own representation space, whereas curvature is defined the same way everywhere, which is the only reason we could say that a reasoning-distilled model shows the same signature as its base models.

We close with two standards for work of this kind. Any trajectory-based interpretability claim should survive two independent controls for text length, because length imitates geometric signal at correlations up to 0.99. And any proposed trajectory method for detection should be required to beat a linear probe on a single pooled hidden state, because in our nine runs that baseline was never beaten.

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A Protocol details

Label rules. For OpenBookQA a chain is valid if a strict majority of crowd annotations equal “yes”, with ties counted invalid. The historical rule, that an annotation contains “yes”, coincides on this corpus and is reported as a sensitivity check. For EntailmentBank we take task-1 and task-2 gold trees with at least three materialized steps as valid, and pair each with one corruption. The corruptions are step shuffle with a guaranteed non-identity permutation, where chains with fewer than three steps escalate to premise substitution, premise substitution from another tree, and conclusion substitution, assigned round-robin with random seed 0. For PrOntoQA we take gold chains of at least three steps from 3-hop generated corpora and corrupt them identically.

Feature list. Token and layer trajectories each contribute path length S , Dirichlet energy E , straightness σ , curvature mean, maximum and standard deviation, Menger circumradius curvature mean as a sensitivity variant, torsion $|\tau|$ mean, standard deviation and defined-fraction, and the phase-space summaries H_{mean} , H_{std} and $|H_{\text{drift}}|$ with kinetic term $\frac{1}{2}\|\Delta_i\|^2$ and alignment potential $-\langle x_i, \bar{q} \rangle / \|\bar{q}\|$, where $\bar{q}$ is the pooled question representation. The uncertainty family contributes NLL, next-token entropy mean and maximum, attention entropy at ℓ^* , and effective rank. The twenty-one geometric features entering the classifiers are those defined on at least 99 per cent of chains in every run, which excludes the torsion defined-fraction summaries and the token-trajectory Menger variant.

Degeneracy guards. Curvature at i is undefined if $\min(\|\Delta_{i-1}\|, \|\Delta_i\|) < \varepsilon$ with $\varepsilon = 10^{-8}$. Torsion is additionally undefined if $\|v_1 \wedge v_2\|^2 \leq 10^{-12} \|v_1\|^2 \|v_2\|^2$. Undefined values are excluded and their fraction reported. Trajectories with $n < 3$ yield empty curvature sets and those with $n < 4$ yield empty torsion sets by construction, so the two-point failure mode of naive pipelines is unrepresentable.

Analytic validation. The released test set checks, prior to any extraction, straight lines with $\kappa \equiv 0$ and $\sigma = 1$, circles of radius r under random high-dimensional isometric embedding with $\hat{\kappa} = 1/r \pm 10^{-3}$ for both (1) and the Menger variant, helices with $\hat{\kappa}$ and $|\widehat{\tau}|$ within 10^{-3} of $r/(r^2+c^2)$ and $c/(r^2+c^2)$, isometry invariance to 10^{-6} , and correct NaN behaviour on degenerate inputs.

B Lean 4 formalization

The development (`Estimators.lean`, compiled with `mathlib` on Lean 4 [Moura and Ullrich, 2021]) defines the estimators over an arbitrary real inner-product space and proves Proposition `efprop.lean`. The core statements are the following.

```
noncomputable def pathLength (n : ℕ) (x : ℕ → E) : ℝ :=
  ∑ i ∈ range n, dist (x i) (x (i + 1))

theorem chord_le_pathLength (n : ℕ) (x : ℕ → E) :
  dist (x 0) (x n) ≤ pathLength n x :=
  dist_le_range_sum_dist x n

theorem pathLength_isometry (n : ℕ) (x : ℕ → E) (f : E ≃ E) :
  pathLength n (fun i => f (x i)) = pathLength n x := by
  unfold pathLength
  exact Finset.sum_congr rfl fun i _ => f.isometry.dist_eq _ _

theorem turningAngle_linearIsometry (x : ℕ → E) (i : ℕ)
  (f : E ≃ E) (t : E) :
  turningAngle (fun j => f (x j) + t) i = turningAngle x i := by
  -- proof in the released file
```

The chord bound reduces to `mathlib’s dist_le_range_sum_dist`, isometry invariance to `Isometry.dist_eq`, and angle invariance to preservation of inner products and norms by linear isometries.

Why a machine-checked proof. The numerical tests of Appendix A establish correctness on the finitely many inputs we sample. A proof on paper covers all inputs but relies on the author and the reader to notice a skipped step, and the step that gets skipped is usually the one that looks obvious. Lean checks every step against a machine that accepts nothing left unstated, which forces the degenerate cases into the open. Straightness is the concrete example. A trajectory that never moves has $S(X) = 0$ and an undefined quotient, so the formalization is obliged to declare a value. We set it to zero, and part (ii) of Proposition 1 is proved with that convention in force, which is exactly the guard the implementation uses. On paper this case usually goes unmentioned and in code it becomes a division by zero.

Scope. The formalization covers definitions and structural properties of the estimators only. It does not formalize the statistical protocol, and it makes no claim about neural networks. In particular, no Noether-type conservation law and no symplectic structure is asserted anywhere in this paper. Noether’s theorem converts each continuous symmetry into a conserved quantity, but only for systems whose trajectories derive from a least-action principle preserved by the symmetry, and nothing establishes that transformer hidden states evolve under such a structure. Symmetry alone yields invariants, which we prove. It yields conservation laws only under variational dynamics, which no one has demonstrated for these systems.

C Complete per-run results

Table 2: Static probe against trajectory geometry, all nine runs, out-of-fold AUROC. “Probe” is the better of the logistic and mass-mean probes on one pooled state. “+geo” is the learned stacker. Δ and p are from the DeLong test of the stack against the probe.

Dataset	Model	Probe	Geometry	Probe+geo	Δ AUROC	p
OpenBookQA	Qwen2.5-7B	0.818	0.735	0.811	−0.007	$< 10^{-4}$
	DeepSeek-R1-8B	0.794	0.771	0.798	+0.004	0.832
	Mistral-7B	0.766	0.744	0.778	+0.012	0.191
EntailmentBank	Qwen2.5-7B	0.862	0.658	0.858	−0.003	0.082
	DeepSeek-R1-8B	0.735	0.649	0.740	+0.005	0.259
	Mistral-7B	0.710	0.662	0.719	+0.009	0.125
PrOntoQA	Qwen2.5-7B	0.989	0.602	0.988	−0.001	0.060
	DeepSeek-R1-8B	0.990	0.659	0.989	−0.001	0.038
	Mistral-7B	0.977	0.745	0.976	−0.001	0.521

Complete per-feature tables, raw, length-residualized and stratified, across 27 CSV files, together with the corruption-type breakdown underlying Figure 3 and the layer-sweep robustness results, are included in the released archive.

D Reproducibility

Every number in this paper is produced by two notebooks and is traceable to a single machine-generated results file (`metrics.json`). Notebook 1 handles extraction. It embeds the estimator library and its analytic test suite, which must pass before any extraction, and emits one feature table and one pooled-state archive per model and dataset pair, nine of each. Notebook 2 handles analysis. It consumes only those artifacts and regenerates all statistics, tables and figures

Table 3: Geometry against the uncertainty family, out-of-fold AUROC, logistic regression. Δ is combined minus uncertainty-only and p is from the DeLong test.

Dataset	Model	Uncertainty	Combined	Δ AUROC	p
OpenBookQA	Qwen2.5-7B	0.782	0.778	-0.004	0.705
	DeepSeek-R1-8B	0.758	0.794	+0.036	0.048
	Mistral-7B	0.762	0.768	+0.006	0.674
EntailmentBank	Qwen2.5-7B	0.714	0.736	+0.022	0.009
	DeepSeek-R1-8B	0.688	0.716	+0.027	0.003
	Mistral-7B	0.694	0.720	+0.026	0.008
PrOntoQA	Qwen2.5-7B	0.811	0.869	+0.058	< 0.01
	DeepSeek-R1-8B	0.675	0.799	+0.124	< 0.01
	Mistral-7B	0.722	0.830	+0.108	< 0.01

Table 4: Length-adjusted effect sizes, Cohen’s d for valid minus invalid, for layer curvature standard deviation and layer torsion. Bold entries are significant under both length controls with BH-FDR $q < 0.05$.

Dataset	Model	Layer curvature	Layer torsion
OpenBookQA	Qwen2.5-7B	-0.69	-0.15
	DeepSeek-R1-8B	-0.62	-0.62
	Mistral-7B	-0.81	-0.85
EntailmentBank	Qwen2.5-7B	-0.38	-0.36
	DeepSeek-R1-8B	-0.31	-0.50
	Mistral-7B	-0.52	-0.52
PrOntoQA	Qwen2.5-7B	+0.01	-0.14
	DeepSeek-R1-8B	-0.28	-0.22
	Mistral-7B	-0.67	-0.63

deterministically with fixed seeds throughout. The released archive contains both notebooks, the estimator and analysis libraries with unit tests, the three chain corpora with their seed-fixed construction script, all eighteen extraction artifacts, all result CSV files, `metrics.json` and `Estimators.lean`.

Models were run in bfloat16. We note that float16 silently corrupts Qwen2.5 hidden states, overflowing to NaN in late layers, a failure mode our NaN accounting caught and which practitioners should be aware of.

E Glossary

This appendix defines every technical term used in the paper. Each entry gives a plain statement of what the object is, followed by the precise definition where one is needed. Readers familiar with one of the three areas the paper draws on, which are curve geometry, transformer interpretability and applied statistics, may wish to read only the other two.

E.1 Trajectories and curve geometry

Real inner-product space

A vector space over the real numbers equipped with an inner product $\langle u, v \rangle$, which is a symmetric bilinear form that is positive on nonzero vectors. The inner product determines a norm $\|u\| = \sqrt{\langle u, u \rangle}$, a distance $\text{dist}(u, v) = \|u - v\|$, and an angle between any two nonzero vectors.

Every hidden-state space $\mathbb{R}^h$ with the usual dot product is an example. All our estimators are defined in this generality, which is why they do not depend on the dimension h .

Trajectory

A finite ordered list of points $X = (x_1, \dots, x_n)$ in such a space. We use two kinds. The token trajectory follows the representation of the text as it advances through the chain at a fixed layer. The layer trajectory follows the representation of the whole chain as it is refined from the first layer to the last.

Step

The vector $\Delta_i = x_{i+1} - x_i$ joining one point of a trajectory to the next. Every quantity in this paper is built from the steps and from inner products between them.

Path length $S(X)$

The total distance travelled, $S(X) = \sum_i \|\Delta_i\|$. It is the discrete counterpart of the arc length of a smooth curve. A path that revisits the same region many times has a large path length even when its endpoints are close together.

Dirichlet energy $E(X)$

The quantity $E(X) = \frac{1}{2} \sum_i \|\Delta_i\|^2$. It measures the same path as S but weights each step by its square, so one large jump contributes more than several small jumps covering the same distance. For a fixed number of steps and a fixed total distance, the energy is smallest when all steps have equal length, which is why it is read as a measure of how evenly a path is traversed.

Chord

The straight segment joining the first and last points of a trajectory. Its length is $\|x_n - x_1\|$.

Straightness $\sigma(X)$

The ratio of the chord to the path length, $\sigma(X) = \|x_n - x_1\|/S(X)$. It compares the distance that separates the endpoints with the distance actually travelled between them. A trajectory that moves in a single direction has $\sigma = 1$. A trajectory that wanders and returns close to its starting point has σ near zero. The quantity is scale-free, meaning that multiplying every point by a constant leaves it unchanged. When $S(X) = 0$ the ratio is undefined and we assign the value zero, which is the convention proved alongside the bound in Proposition 1 and the guard used in the implementation.

Turning angle θ_i

The angle between the step arriving at a point and the step leaving it, $\theta_i = \arccos(\langle \Delta_{i-1}, \Delta_i \rangle / (\|\Delta_{i-1}\| \|\Delta_i\|))$. It lies between 0 and π . A value of zero means the path continues straight ahead and a value of π means it doubles back.

Discrete curvature κ_i

The turning angle divided by the mean of the two adjacent step lengths, as in (1). Dividing by a length converts a pure angle into a turn per unit of distance travelled, which is the quantity that converges to the curvature of a smooth curve as the sampling becomes dense. Without that division, a path sampled at twice the resolution would appear to turn half as sharply at each point.

Menger curvature

An alternative discrete curvature obtained from three consecutive points by taking the reciprocal of the radius of the unique circle through them. We report it as a sensitivity variant, so that no conclusion depends on one particular discretization of curvature.

Osculating plane

The plane that best contains the curve near a given point, spanned by the first two derivative estimates. A path that stays inside one plane bends but does not twist.

Torsion $|\tau_i|$

The rate at which a path leaves its own osculating plane. A road bending on a flat map has curvature and no torsion, whereas a spiral staircase has both. We estimate it from four consecutive points using the finite-difference approximations v_1, v_2, v_3 of the first three derivatives, as in (2). We report the unsigned value, so a left-handed and a right-handed twist of equal magnitude receive the same score.

Gram matrix G

The matrix whose entry in row j and column k is $\langle v_j, v_k \rangle$. Its determinant equals the squared volume of the parallelepiped spanned by the vectors. That determinant vanishes exactly when the vectors are linearly dependent, which for v_1, v_2, v_3 means the path is locally flat, and this is why the torsion estimate is zero for planar paths by construction rather than by numerical accident.

Isometry

A map of the ambient space to itself that preserves distances, so that $\text{dist}(f(u), f(v)) = \text{dist}(u, v)$ for all u and v . In a real inner-product space every isometry is a linear isometry followed by a translation.

Linear isometry

An isometry that is also linear, therefore fixing the origin and preserving inner products as well as distances. Rotations and reflections are linear isometries. Translations are isometries but are not linear.

Isometry group

The set of all isometries of the space, which is closed under composition and inversion. Its elements are the changes of description that leave all distances and angles intact.

Invariant

A quantity that takes the same value before and after every transformation in a given group. Proposition 1 states that S , E and σ are invariants of the full isometry group and that turning angles are invariants of the subgroup generated by linear isometries and translations.

Erlangen programme

Klein’s proposal that a geometry is determined by the group of transformations under which its quantities are invariant, rather than by the space itself. Under this view, exhibiting our estimators as invariants of the isometry group is what entitles us to call the analysis geometric.

E.2 Model internals**Hidden state**

The vector in $\mathbb{R}^h$ that the model holds at one token position and one layer. It is the object all our measurements read.

Residual stream

The running vector that each transformer layer reads from and adds to. Because layers write increments rather than replacements, the sequence of residual stream values across depth is naturally read as a path, which is what makes the layer trajectory well defined.

Layer

One block of the transformer. Depth index ℓ runs from the embedding at $\ell = 0$ to the final block at $\ell = L$.

Mean pooling

Averaging the hidden states of all token positions into a single vector. We use it to reduce each layer to one point of the layer trajectory, and to produce the single vector the probe reads.

Pre-specified layer ℓ^*

The depth at which token-level features and the probe are extracted, fixed before the analysis so that no result depends on a layer chosen after seeing the outcome. Sweeps over other layers are reported as robustness checks.

Reasoning chain

A sequence of natural-language steps leading to a conclusion. Validity is a property of the sequence and not of any single step, which is the distinction that motivates measuring paths rather than points.

Corruption

A controlled edit that turns a valid chain into an invalid one. We use three. Shuffling reorders the steps and leaves every sentence intact. Premise substitution replaces one step with a sentence taken from a different problem. Conclusion substitution replaces the final step. Because the edit is known for every invalid chain, each detector can be scored separately against each kind of failure.

Effective rank

A continuous measure of how many directions a set of vectors genuinely occupies. It is the exponential of the Shannon entropy of the singular values of the hidden-state matrix after normalizing them to sum to one. A set concentrated along one direction has effective rank near one, and a set spreading its variance evenly across k directions has effective rank near k .

E.3 Detection and probing**Probe**

A simple classifier trained on frozen internal activations rather than on text. It tests whether a property is present in a representation in a form a simple reader can extract, and it does not modify the model.

Linear probe

A probe whose decision rule is a single direction in representation space together with a threshold. Finding that a property is linearly readable is a stronger statement than finding it readable, because a linear rule can only exploit information already aligned with one direction.

Logistic regression probe

A linear probe whose direction is fitted by maximizing the likelihood of the labels.

Mass-mean probe

A linear probe whose direction is the vector from the mean of the negative class to the mean of the positive class, optionally corrected by the inverse covariance of the centred data. It requires no optimization. We use the better of this and the logistic probe as our baseline, following the practice established for single-statement truth.

Detector family

A group of features evaluated as one method. We compare three, which are the geometric features of §3, the uncertainty features derived from the model’s own confidence, and the probe on one pooled hidden state.

Dimension-matched baseline

A probe restricted to the same number of dimensions as the competing feature set, obtained here by random projection or by principal component analysis of the pooled state. It separates the question of whether the probe has more information from the question of whether it has more capacity.

Depth-concatenated probe

A linear probe trained on the per-layer pooled states joined into one long vector, so that the linear reader has the same access to depth as a trajectory-based method. It is the matched baseline for any claim that a depth-based method adds nothing.

Stacking

Training a second classifier on the out-of-fold predictions of several base detectors, so that the combination can weight a strong detector above a weak one instead of averaging them. We use it so that no family is penalized by being combined with a weaker one.

Cross-validation

Splitting the data into folds, training on all but one and evaluating on the one held out, then rotating. It estimates performance on data the classifier has not seen. We use five stratified folds repeated ten times with fixed seeds.

Out-of-fold prediction

A prediction made for a sample by a model that did not see that sample during training. All reported scores are computed from out-of-fold predictions.

Nuisance covariate

A variable included in a model not because it is of interest but so that the remaining variables cannot earn credit for it. We include chain length in every classifier for this reason.

Negative control

A run of the identical pipeline on data where no true effect can exist. Ours uses features extracted from a model with random weights, and it correctly returns chance performance and no significant features, which shows the pipeline does not manufacture signal.

E.4 Statistical terms**AUROC**

The area under the receiver operating characteristic curve. It equals the probability that a randomly chosen valid chain receives a higher score than a randomly chosen invalid one. A value of 0.5 is chance and a value of 1 is perfect separation. It does not depend on any threshold and it is insensitive to class imbalance, which matters here because the OpenBookQA base rate is 90.5 per cent and raw accuracy would therefore be uninformative.

Balanced accuracy

The mean of the accuracy on the positive class and the accuracy on the negative class, which equals ordinary accuracy when the classes are balanced.

Matthews correlation coefficient

The correlation between predicted and true binary labels, ranging from -1 to 1 , with 0 at chance. It falls when a classifier succeeds on only one class, which is why we report it alongside AUROC.

Cohen’s d

An effect size given by the difference between two group means divided by their pooled standard deviation. It expresses a difference in units of the spread of the data, so it can be compared across features measured on different scales. Values near 0.2 are conventionally small and values near 0.8 are conventionally large.

Length-adjusted effect size

Cohen’s d computed on residualized features rather than raw ones, so that the reported difference is the part not attributable to chain length.

Cliff’s δ

A rank-based effect size equal to the probability that a value drawn from one group exceeds a

value drawn from the other, minus the probability of the reverse. It ranges from -1 to 1 and assumes nothing about the shape of the distributions.

Welch's t test

A test for a difference between two means that does not assume the two groups have equal variance.

Mann–Whitney U test

A test of whether values from one group tend to be larger than values from the other, based only on ranks. It makes no distributional assumption, and its normalized statistic coincides with AUROC, so the per-feature tests and the detection metric measure separation in the same sense.

p value

The probability of observing a statistic at least as extreme as the one obtained, under the assumption that no effect is present.

Multiple comparisons

The problem that testing many features raises the chance that some cross a significance threshold by accident.

Benjamini–Hochberg FDR and q

A correction that controls the false discovery rate, meaning the expected proportion of false positives among the results declared significant. We report q values and use $q < 0.05$. This differs from a Bonferroni correction, which controls the probability of even one false positive and is far more conservative when many features are tested.

Bootstrap confidence interval

An interval obtained by resampling the data with replacement many times and recomputing the statistic each time. We use ten thousand resamples. It requires no assumption that the statistic is normally distributed.

DeLong test

A test for the difference between two AUROC values computed on the same samples. Using it rather than an independent-sample test matters here because our detectors score the same chains, so their errors are correlated.

Permutation test

A test that builds a null distribution directly by repeatedly shuffling the labels and recomputing the statistic. It assumes only that the labels are exchangeable under the null.

Residualization

Regressing a feature on a set of predictors and keeping only the part it does not explain. We regress on the basis $\{1, n, n^2, \log n\}$ of functions of chain length, so the residual contains no component expressible as a smooth function of length within that span. This is stronger than removing a straight-line trend.

Stratification

Dividing the data into groups that are homogeneous in a nuisance variable and testing within each. We use length deciles, so every comparison is made between chains of similar length.

Stouffer combination

A method for pooling the results of several independent tests by averaging their z scores and rescaling by the square root of their number. We use it to combine the within-decile tests into one statement.

Confound

A variable that influences both the measurement and the label, so that an association between

them can arise without any direct relation. Chain length is the confound this paper takes most seriously, because valid chains are simply shorter in OpenBookQA and length drives several geometric features at correlations up to 0.99.

Dissociation

A pattern in which two measurements respond differently to two experimental manipulations. Our geometric detector responds to corrupted content and remains at chance on corrupted order, whereas the probe responds to both, and this contrast is what licenses the claim that trajectory shape reads content rather than structure.

E.5 Formal verification

Proof assistant

A program in which mathematical definitions and proofs are written in a formal language and every inference is checked mechanically. It does not discover proofs. It refuses to accept an argument with a gap.

Lean 4

The proof assistant used here. Its correctness rests on a small trusted kernel that rechecks every completed proof, so an error elsewhere in the system cannot by itself make a false statement provable.

mathlib

The community library of formalized mathematics for Lean, which supplies the definitions of metric spaces, inner-product spaces and isometries that our development builds on. Our proof of the chord bound, for instance, is an application of the library result `dist_le_range_sum.dist`.

Formalization

The act of restating a definition or theorem in the formal language of a proof assistant. The restatement is itself a claim, since a formalization can be faithful to the informal statement or not, and the machine checks the proof rather than the faithfulness.

Machine-checked

Verified by such a system. The guarantee is that the stated theorem follows from the stated definitions. It is not a guarantee that the definitions are the ones intended, nor that the numerical implementation agrees with them, which is why we also run the numerical validation of Appendix A.

Total function

A function required to return a value on every input. Lean requires totality, so a definition involving division must say what happens when the denominator is zero. This requirement is what forces the degenerate case of straightness into the open rather than leaving it to be discovered at run time.

Junk value

The value assigned in a case where no meaningful answer exists, chosen so that a definition is total. We set $\sigma(X) = 0$ when $S(X) = 0$, and Proposition 1 is proved with that convention in force.

IsometryEquiv, written $\simeq_i$

The mathlib type of distance-preserving bijections of a space to itself. It is the formal object corresponding to the phrase every isometry of the ambient space in part (iii) of Proposition 1.

LinearIsometryEquiv, written $\simeq_{\ell_i}$

The mathlib type of linear isometric bijections. It is the formal object used for the turning-angle statement, where linearity is needed because angles are computed from difference vectors and

inner products rather than from distances alone.